

AI Research Projects in Collaboration with Sony Interactive Entertainment (SIE) and the DICE Lab at Royal Holloway

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Introduction

The following projects investigate cutting-edge applications of machine learning and AI to video game development workflows, with a particular focus on assisting Quality Assurance (QA) engineers. Students participating in these projects will have opportunities to interact with experienced ML researchers at SIE, and some may be considered for placements or internships. Please note that both projects include an additional application step involving a short interview with SIE researchers.

Project 1: VLMs for Video Game QA – Relationships between Code and Intended Behaviour

Aims

This project aims to explore the capabilities of (Large) Vision Language Models (VLMs) in understanding the relationship between the behaviour of a game as the player sees it, and the code that implements this behaviour. Specifically, it investigates whether VLMs can identify “regression” bugs, where code or asset changes unintentionally alter game behaviour between versions.

Background

Quality Assurance (QA) is a vital part of video game development. With recent developments in AI, there is growing interest in tools that assist QA engineers with the testing process. Video games present a particularly challenging domain for test automation due to the complexity of interactions between player inputs and audiovisual output, the numerous subsystems involved (physics, graphics, AI, networking, etc.), and the subjective nature of “intended behaviour”.

Recent work applying VLMs to game testing has shown promising, though limited, results—for example, detecting simple bugs such as clipping issues (Taesiri 2025). This project extends such work by introducing game code as an additional input signal, enabling investigation into whether VLMs can explain behavioural changes and help identify potential regression bugs.

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The project is conducted in collaboration with Sony Interactive Entertainment (SIE), offering students opportunities to engage with industry researchers and potentially secure placement opportunities.

Early Deliverables

- Literature review on the use of VLMs and related models in video game testing and automated bug detection.
- Construction of an evaluation dataset containing code/asset diffs and gameplay images or videos, generated from an open-source 3D game engine project, including both intended and unintended changes.
- Initial evaluation of a chosen VLM's ability to caption or describe the generated scenes.

Final Deliverables

- Evaluation of whether VLMs can make effective use of code diffs to identify behavioural changes in scenes.
- Evaluation of VLMs' ability to detect regression bugs with and without code context.
- A report detailing the literature review, design decisions (e.g., prompt engineering), experimental methodology, and results.

Possible Extensions

- Comparative study of open- and closed-source VLMs applied to this task.

Indicative Reading

- Taesiri, Mohammad Reza et al. "VideoGameQA-Bench: Evaluating Vision-Language Models for Video Game Quality Assurance". arXiv, 2025.
<https://doi.org/10.48550/ARXIV.2505.15952>
- Cheng, An-Chieh et al. "SpatialRGPT: Grounded Spatial Reasoning in Vision-Language Models". NeurIPS 2024.
<https://openreview.net/forum?id=JKEIYQUSUc>
- Zhang, Boran et al. "Human-AI Collaborative Game Testing with Vision Language Models". arXiv:2501.11782, 2025.
<https://doi.org/10.48550/arXiv.2501.11782>

Prerequisites

- Strong understanding of machine learning fundamentals.
- Experience with generative AI, including API usage and prompt engineering; familiarity with local model execution and fine-tuning is beneficial.

- Strong Python skills and experience with scripting in at least one 3D game engine (Unity, Unreal, Godot, etc.).
- Experience with git.
- Ability to work independently and solve complex problems.
- Interest in video games.

Project 2: Detecting Technical Requirement Violations in Video Games with (Large) Language Models

Aims

The aim of this project is to develop a machine-learning-based pipeline to assist QA engineers in ensuring that video games adhere to platform-specific technical requirements documents.

Background

Gaming platforms impose strict requirements on published titles to ensure a consistent player experience and compliance with legal obligations. Requirements include correct terminology for platform features, copyright and trademark compliance, and safeguarding player privacy and safety. Currently, evaluation against Technical Requirements Checklists (TRCs) is performed manually, making the process slow and costly.

With recent advances in NLP, particularly LLMs and contextual embedding models, automated approaches to compliance checking have begun to show promise in other domains such as legal text analysis (Hassani 2024). This project explores similar approaches for analysing text extracted from video game screenshots to identify TRC violations.

As with the previous project, this work is conducted in collaboration with SIE, and involves an additional interview stage for applicants.

Early Deliverables

- Literature review of LLM/RAG approaches to software and video game requirement checking.
- Evaluation of existing embedding models or open-source LLMs on TRC-related tasks.
- A trained or fine-tuned model capable of reliably detecting a subset of TRC violations from text extracted from real game screenshots.

Final Deliverables

- A pipeline that ingests a screenshot, extracts text, and identifies a subset of TRC violations across major world languages.
- Robust evaluation of the pipeline, including ablation studies where appropriate and comparison to alternative methods.
- A report presenting the literature review, design rationale, and detailed experimental results.

Possible Extensions

- A model capable of detecting nuanced TRC violations requiring reasoning or broader contextual understanding, such as those relating to player safety or privacy.

Indicative Reading

- Hassani, Shabnam et al. “Rethinking Legal Compliance Automation: Opportunities with Large Language Models”. arXiv:2404.14356, 2024.
- Nie, Zhijie et al. “When Text Embedding Meets Large Language Model: A Comprehensive Survey”. arXiv:2412.09165, 2025.
- Zhang, Xin et al. “mGTE: Generalized Long-Context Text Representation and Reranking Models for Multilingual Text Retrieval”. arXiv:2407.19669, 2024.
See also: <https://huggingface.co/Alibaba-NLP/gte-large-en-v1.5>

Prerequisites

- Strong Python programming skills.
- Proven experience with PyTorch and the wider ML ecosystem (e.g. Hugging Face).
- Working knowledge of embedding models and LLMs.
- Understanding of machine learning and deep learning fundamentals.
- Interest in video games.